Lenin T

Coimbatore, Tamil Nadu

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Summary

ECE graduate with hands-on experience in Python development, backend engineering, machine learning, and a local AI code assistant. Seeking an entry-level Software Engineer or AI/ML Engineer role focused on building practical solutions.

Education

Kumaraguru College of Technology **Nov. 2022 – May 2026**
B.E. in Electronics and Communication Engineering – CGPA: 7.90 *Coimbatore, Tamil Nadu*

Sri Vivekananda Matriculation Higher Secondary School
Class XII – 86.6% — *Class X – 96.2%* *Coimbatore, Tamil Nadu*

Technical Skills

Languages & Core Concepts: Python; SQL; C; Object-Oriented Programming; Data Structures and Algorithms

Software & Web Development: FastAPI; REST APIs; SQLAlchemy; JWT Authentication; React; HTML; CSS

Databases & Data Analysis: PostgreSQL; SQLite; NumPy; Pandas; Matplotlib; Data Preprocessing; Exploratory Data Analysis

Machine Learning & Generative AI: scikit-learn; Random Forest; Model Evaluation; Classification Metrics; PyTorch; Sentence Transformers; Embeddings; FAISS; Python AST parsing; semantic code chunking; RAG; Hybrid Retrieval; Ollama; Prompt Engineering

Tools & Technologies: Selenium; pytest; Git; GitHub; Docker; Linux

Projects

RepoPilot AI – Local Codebase Assistant **Nov. 2026**
Tech Stack: Python, AST, Sentence Transformers, FAISS, Ollama

- Built a CLI code assistant that parses Python repositories, creates semantic chunks, and retrieves relevant source using embeddings, FAISS, and hybrid lexical reranking.
- Added a provider-independent RAG pipeline with local Ollama generation; verified with 69 unit tests and an 8-question retrieval evaluation.

ToLet Coimbatore – Hyperlocal Rental Platform **Apr. 2026 – Jun. 2026**
Tech Stack: Python, FastAPI, React, PostgreSQL, SQLAlchemy, JWT, Cloudinary, Railway

- Built a full-stack rental platform with a React frontend and a FastAPI/PostgreSQL backend, supporting listings, JWT authentication, and Cloudinary photo uploads.
- Deployed the application on Railway with a cloud PostgreSQL database.

PredictIQ – Predictive Maintenance System **Oct. 2026**
Tech Stack: Python, scikit-learn, FastAPI, SQLite, Docker

- Built a Random Forest model using the AI4I 2020 dataset to predict machine failures; the model achieved 98.15% test accuracy.
- Served predictions through FastAPI, stored history in SQLite, and containerized the application with Docker.

Certifications

- Build & Deploy AI Applications with Python & Flask – IBM, Coursera
- Python for Data Science, AI & Development – IBM, Coursera
- Generative AI: Introduction and Applications – IBM, Coursera
- Generative AI: Prompt Engineering Basics – IBM, Coursera